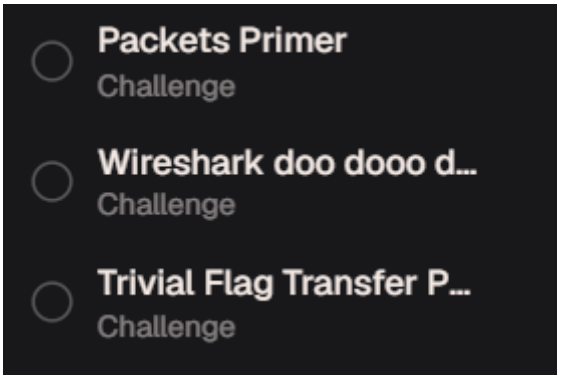
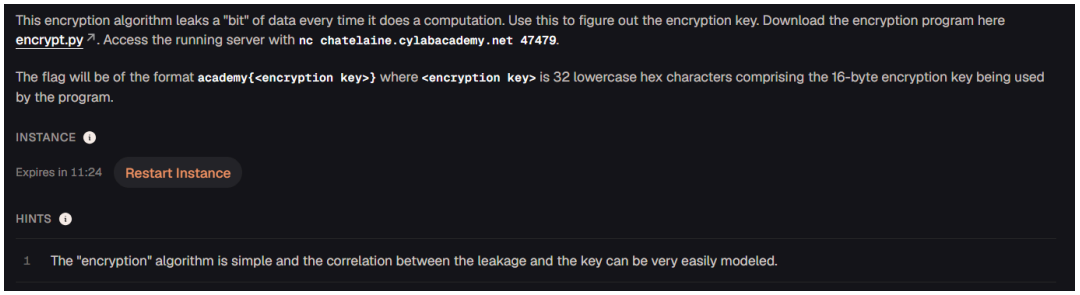


Challenge 1 : Packets Primer



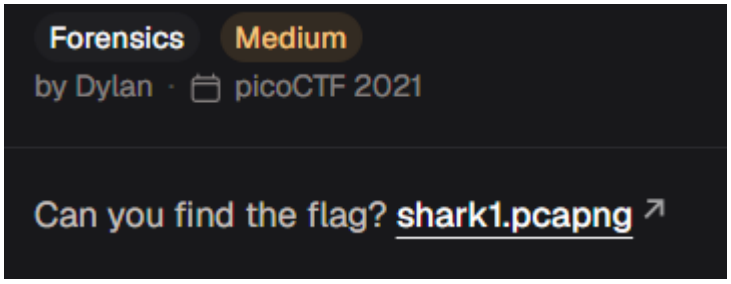
File Bernama : network-dump.flag.pcap

1. Kita cek filenya dengan exiftool seperti biasa.
2. Kita coba baca isi filenya dengan **cat network-dump.flag.pcap** dan yap ada yang janggal



3. Tinggal kita rapiin dengan **`echo "" | tr -d ' '`** dan ketemu flagnya yakni `academy{p4ck37_5h4rk_1c872e64}`

Challenge 2 : Wireshark doo dooo do doo...



File bernama : shark1.pcapng

1. Kita cek informasi filenya dengan exiftool
2. Kita coba cari lewat strings dan tidak ketemu apa apa.
3. Coba dengan library tshark dengan query : **tshark -r shark1.pcapng -q -z io,phs**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ tshark -r shark1.pcapng -q -z io,phs

===================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:987 bytes:1524814
  eth                                    frames:987 bytes:1524814
    ip                                   frames:985 bytes:1524716
      tcp                                frames:985 bytes:1524716
        http                             frames:288 bytes:1158734
          mime_multipart                 frames:142 bytes:78526
          data-text-lines                frames:2 bytes:695
        tls                              frames:43 bytes:75441
    arp                                  frames:2 bytes:98
===================================================================
```

4. Dari output diketahui bahwa http memiliki hal yang curiga yakni ukuran bytes 1.1 MB dan file **shark1.pcapng** seluruhnya memiliki 1.5 MB. Alasan lainnya ada disini penjelasannya : https://chat.deepseek.com/share/6w14tuoyz867x42rn4
5. Lalu kita **Extract HTTP Objects dengan tshark** dengan 3 query berurutan ini

```
mkdir http_objects
tshark -r shark1.pcapng --export-objects http,http_objects
ls -la http_objects/
```

6. Ada file janggal seperti 36, 36(1) - 36(139), 37, 43, %2f, instance-action. jadinya kita melakukan pengecekkan satu satu filenya. Ada sesuatu yang janggal pada output **file http_objects/%2f** berupa outputnya kode ASCII

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ file http_objects/36
file http_objects/37
file http_objects/43
file http_objects/%2f
http_objects/36: data
http_objects/37: data
http_objects/43: data
http_objects/%2f: ASCII text
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cat http_objects/%2f:
cat: http_objects/%2f:: No such file or directory
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cat http_objects/%2f
Gur synt vf npnqrzl{c33xno00_1_f33_h_qrnqorrs}
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ file http_objects/36
file http_objects/37
file http_objects/43
file http_objects/%2f
http_objects/36: data
http_objects/37: data
http_objects/43: data
http_objects/%2f: ASCII text
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cat http_objects/%2f
Gur synt vf npnqrzl{c33xno00_1_f33_h_qrnqorrs}
```

7. Selanjutnya tinggal kita decode kode ASCII dan yap ketemu flagnya yakni `academy{p33kab00_1_s33_u_deadbeef}`

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ echo "Gur synt vf npnqrzl{c33xno00_1_f33_h_qrnqorrs}"
| tr 'A-Za-z' 'N-ZA-Mn-za-m'
The flag is academy{p33kab00_1_s33_u_deadbeef}
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$
```

Challenge 3 : Trivial Flag Transfer Protocol



File bernama : trfp.pcapng

1. Kita cek info file dengan exiftool, lalu cek protokol dengan tshark -r tftp.pcapng -q -z io,phs output yang keluar ternyata tftp yang digunakan bukan http
2. Lihat file yang ditransfer dengan tshark -r tftp.pcapng -Y "tftp" -T fields -e tftp.source_file -e tftp.destination_file | sort -u

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ tshark -r tftp.pcapng -Y "tftp" -T fields -e tftp.sour
ce_file -e tftp.destination_file | sort -u
        instructions.txt
        picture1.bmp
        picture2.bmp
        picture3.bmp
        plan
        program.deb
picture1.bmp
picture2.bmp
picture3.bmp
plan
program.deb
```

3. Buat file solve bertipe python (**nano solve.py**). Source Code ada di file bernama **solve.py**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ tshark -r tftp.pcapng -Y "tftp.opcode == 3" -T fields
-e udp.payload | head -5", "udp.srcport", "-e", "udp.dstport", "-e", "udp.payload"],


lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ nano solve
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ nano solve.py
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ python3 solve.py
Saved extracted_34790_52355.bin (113 bytes)
Saved extracted_41165_44157.bin (59 bytes)
Saved extracted_37790_37326.bin (138310 bytes)
Saved extracted_33073_59251.bin (824518 bytes)
Saved extracted_38171_44635.bin (36578358 bytes)
Saved extracted_33810_49871.bin (1466574 bytes)
```

4. Cek tipe tiap file dan ada yang janggal pada file extracted_41165_44157 dan extracted_34790_52355 karena tertulis ascii text. Lalu kita baca file janggal tersebut dengan cat dan lakukan decode terhadap file janggal tersebut.

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ file extracted_*.bin
extracted_33073_59251.bin: PC bitmap, Windows 3.x format, 605 x 454 x 24, image size 824464, resolution 5669 x 5669 px/m, cbSize 82451
8, bits offset 54
extracted_33810_49871.bin: PC bitmap, Windows 3.x format, 807 x 605 x 24, image size 1466520, resolution 5669 x 5669 px/m, cbSize 1466
574, bits offset 54
extracted_34790_52355.bin: ASCII text
extracted_37790_37326.bin: Debian binary package (format 2.0), with control.tar.gz , data compression xz
extracted_38171_44635.bin: PC bitmap, Windows 3.x format, 4032 x 3024 x 24, image size 36578304, resolution 5669 x 5669 px/m, cbSize 3
6578358, bits offset 54
extracted_41165_44157.bin: ASCII text
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cat extracted_41165_44157.bin
VHFRQGURCEBTENZNAQUVQVGJVGU-QHRQVYVTRAPR.PURPXBHGGURCUBGBF
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ echo "VHFRQGURCEBTENZNAQUVQVGJVGU-QHRQVYVTRAPR.PURPXBH
GGURCUBGBF" | tr 'A-Za-z' 'N-ZA-Mn-za-m'
IUSEDTHEPROGRAMANDHIDITWITH-DUEDILIGENCE.CHECKOUTTHEPHOTOS
```

5. Kita extract program pada file extracted_37790_37326 karena membuat binary package

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cp extracted_37790_37326.bin program.deb
ar x program.deb
ls -la
total 90812
drwxrwxrwx 1 lenovo lenovo     4096 Oct 16 14:41 .
drwxrwxrwx 1 lenovo lenovo     4096 Oct 16 14:30 ..
-rwxrwxrwx 1 lenovo lenovo     1250 Oct 16 14:41 control.tar.gz
-rwxrwxrwx 1 lenovo lenovo   136868 Oct 16 14:41 data.tar.xz
-rwxrwxrwx 1 lenovo lenovo        4 Oct 16 14:41 debian-binary
-rwxrwxrwx 1 lenovo lenovo   824518 Oct 16 14:39 extracted_33073_59251.bin
-rwxrwxrwx 1 lenovo lenovo  1466574 Oct 16 14:39 extracted_33810_49871.bin
-rwxrwxrwx 1 lenovo lenovo      113 Oct 16 14:39 extracted_34790_52355.bin
-rwxrwxrwx 1 lenovo lenovo   138310 Oct 16 14:39 extracted_37790_37326.bin
-rwxrwxrwx 1 lenovo lenovo 36578358 Oct 16 14:39 extracted_38171_44635.bin
-rwxrwxrwx 1 lenovo lenovo       59 Oct 16 14:39 extracted_41165_44157.bin
drwxrwxrwx 1 lenovo lenovo     4096 Oct 16 14:03 http_objects
-rwxrwxrwx 1 lenovo lenovo      778 Oct 16 13:42 network-dump.flag.pcap
-rwxrwxrwx 1 lenovo lenovo   138310 Oct 16 14:41 program.deb
-rwxrwxrwx 1 lenovo lenovo  1558808 Oct 16 13:50 shark1.pcapng
-rwxrwxrwx 1 lenovo lenovo     1159 Oct 16 14:38 solve
-rwxrwxrwx 1 lenovo lenovo     1159 Oct 16 14:39 solve.py
-rwxrwxrwx 1 lenovo lenovo 52116496 Oct 16 14:28 tftp.pcapng
```

6. Setelah kita extract, kita extract lagi file data.tar.xz dan muncul direktori usr

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ tar xf data.tar.xz
ls -la
total 90812
drwxrwxrwx 1 lenovo lenovo     4096 Oct 15  2014 .
drwxrwxrwx 1 lenovo lenovo     4096 Oct 16 14:30 ..
-rwxrwxrwx 1 lenovo lenovo     1250 Oct 16 14:41 control.tar.gz
-rwxrwxrwx 1 lenovo lenovo   136868 Oct 16 14:41 data.tar.xz
-rwxrwxrwx 1 lenovo lenovo        4 Oct 16 14:41 debian-binary
-rwxrwxrwx 1 lenovo lenovo   824518 Oct 16 14:39 extracted_33073_59251.bin
-rwxrwxrwx 1 lenovo lenovo  1466574 Oct 16 14:39 extracted_33810_49871.bin
-rwxrwxrwx 1 lenovo lenovo      113 Oct 16 14:39 extracted_34790_52355.bin
-rwxrwxrwx 1 lenovo lenovo   138310 Oct 16 14:39 extracted_37790_37326.bin
-rwxrwxrwx 1 lenovo lenovo 36578358 Oct 16 14:39 extracted_38171_44635.bin
-rwxrwxrwx 1 lenovo lenovo       59 Oct 16 14:39 extracted_41165_44157.bin
drwxrwxrwx 1 lenovo lenovo     4096 Oct 16 14:03 http_objects
-rwxrwxrwx 1 lenovo lenovo      778 Oct 16 13:42 network-dump.flag.pcap
-rwxrwxrwx 1 lenovo lenovo   138310 Oct 16 14:41 program.deb
-rwxrwxrwx 1 lenovo lenovo  1558808 Oct 16 13:50 shark1.pcapng
-rwxrwxrwx 1 lenovo lenovo     1159 Oct 16 14:38 solve
-rwxrwxrwx 1 lenovo lenovo     1159 Oct 16 14:39 solve.py
-rwxrwxrwx 1 lenovo lenovo 52116496 Oct 16 14:28 tftp.pcapng
drwxrwxrwx 1 lenovo lenovo     4096 Oct 15  2014 usr
```

7. Kita rename 3 file yang berisi file BMP dan **Extract dengan steghide + Password DUEDILIGENCE** di flag.txt untuk menemukan flagnya.

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cp extracted_33073_59251.bin picture1.bmp
cp extracted_33810_49871.bin picture2.bmp
cp extracted_38171_44635.bin picture3.bmp
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ steghide extract -sf picture1.bmp -p DUEDILIGENCE
steghide extract -sf picture2.bmp -p DUEDILIGENCE
steghide extract -sf picture3.bmp -p DUEDILIGENCE
steghide: could not extract any data with that passphrase!
wrote extracted data to "flag.txt".
steghide: could not extract any data with that passphrase!
```

8. Tinggal kita baca filenya dengan cat dan yap ketemu flagnya yakni `academy{h1dd3n_1n_pLa1n_51GHT_18375919}`

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$ cat flag.txt
academy{h1dd3n_1n_pLa1n_51GHT_18375919}
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-4$
```